

Project Submission Documentation

Home Assistant OS (HAOS) Universal Remote Framework

1. Project Component Checklist

- **Working Home Assistant VM Instance:** Fully configured and operational inside a localized Virtual Machine environment.
- **Mock Integration Core:** Deployed successfully via native State Helper Storage Drivers to replicate independent device registers.
- **Unified Dashboard Interface:** Cohesive dual-panel design displaying peripheral control components paired horizontally alongside diagnostic logging entities.

2. System Deployment Screenshots

Below are the captured interface setups showing the final deployment layout and helper assets:

Figure 1: Unified Live Remote Controls & Debug View Panel

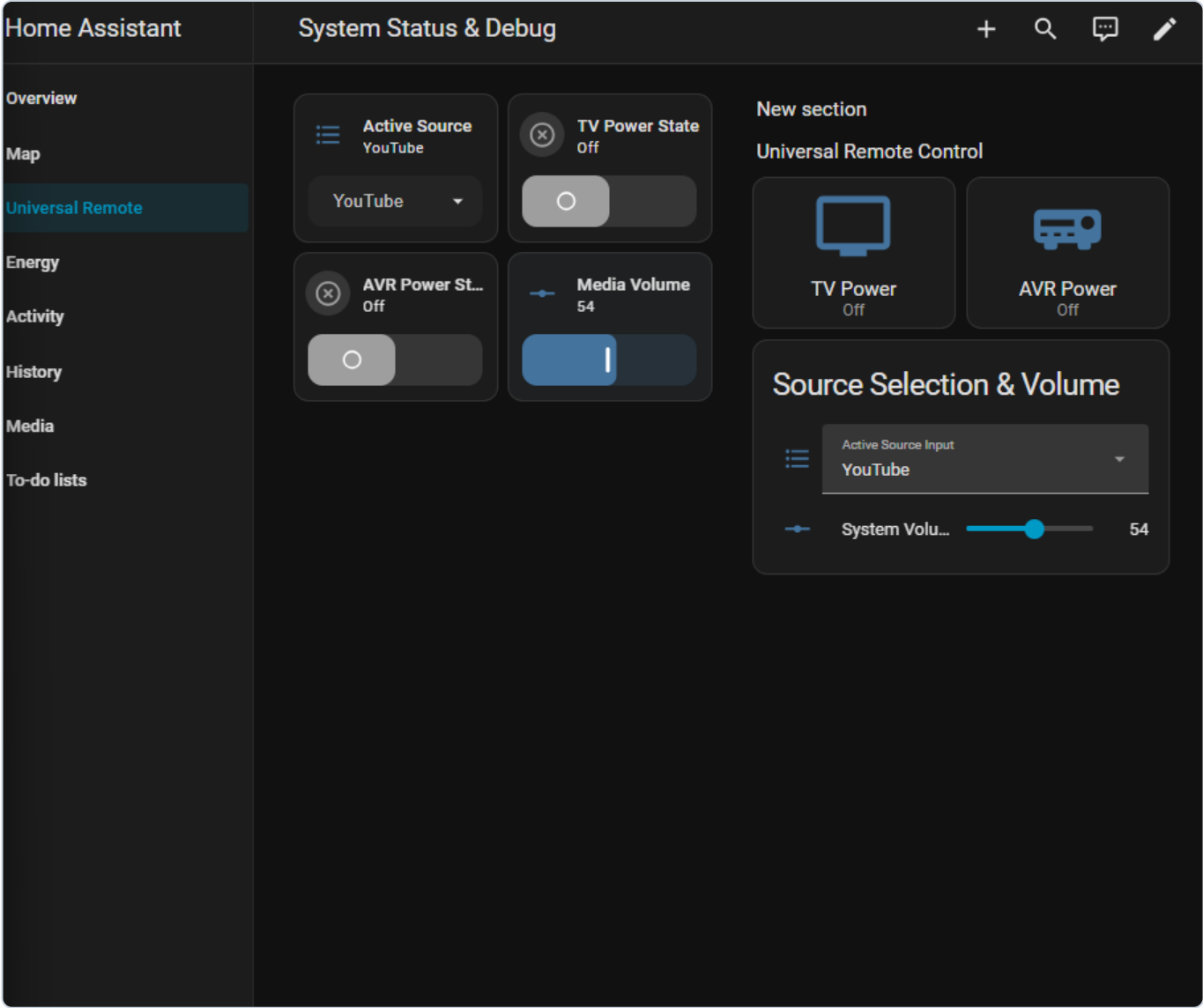
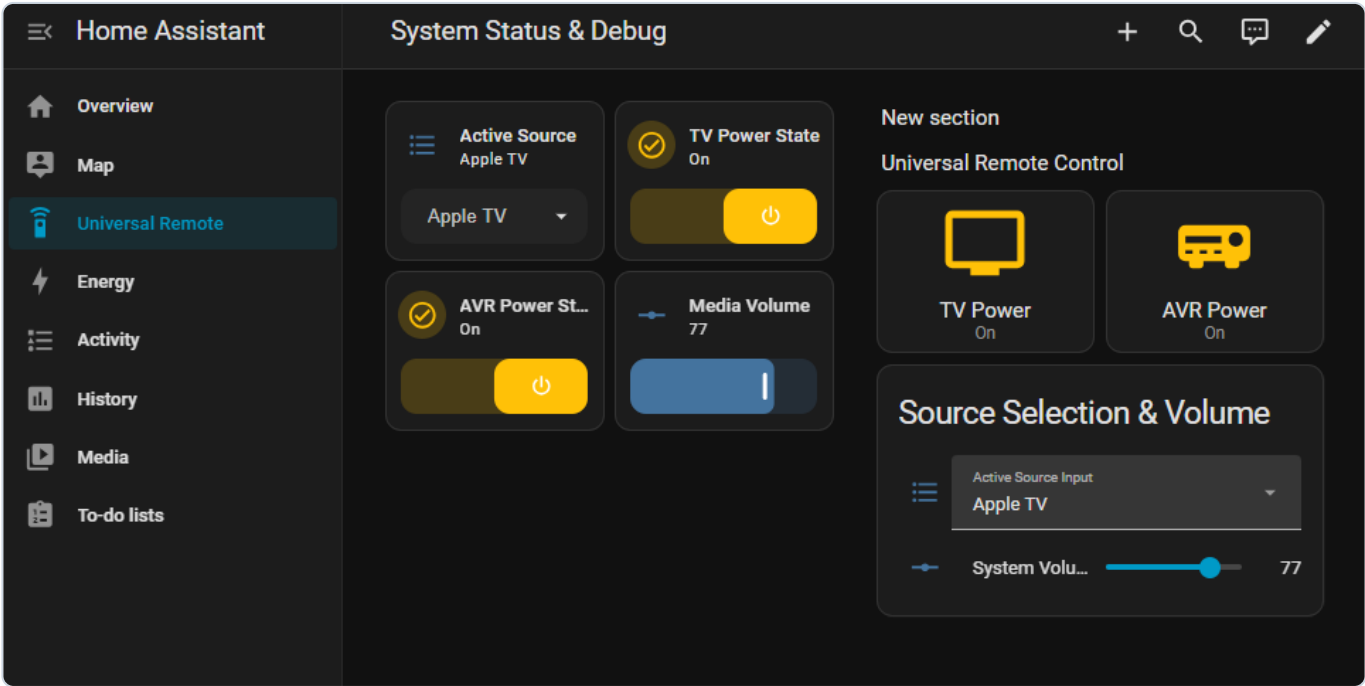


Figure 2: Virtual Helpers System Telemetry Registers Configuration



3. Dashboard Deployment Source File

The layout configuration utilizes a structural card-stack matrix. Use the declarative Lovelace configuration source code below to perfectly instantiate or reproduce the complete functional front-end remote engine setup:

```

type: vertical-stack
cards:
  - type: heading
    heading: Universal Remote Control
  - type: grid
    square: false
    columns: 2
    cards:
      - type: button
        name: TV Power
        entity: input_boolean.tv_power_state
        icon: mdi:television
        show_state: true
        tap_action:
          action: toggle
      - type: button
        name: AVR Power
        entity: input_boolean.avr_power_state
        icon: mdi:audio-video
        show_state: true
        tap_action:
          action: toggle
  - type: entities
    title: Source Selection & Volume
    entities:
      - entity: input_select.active_source
        name: Active Source Input
      - entity: input_number.media_volume
        name: System Volume

```

4. README: Architecture & Configuration Setup

Architecture Overview

The internal solution layout structurally decouples visual dashboard interactive layers from the lower-tier integration variables using abstract virtualization state handlers:

- **Virtual Mock Integration:** Deploys native localized environment registers (`input_boolean` , `input_select` , and `input_number` tracking elements) to emulate complex system states without mapping actual smart hardware components.
- **State Synchronization Core:** A specialized back-end automation observer pipeline detects updates on the tracking elements. It executes service payloads dynamically to prevent state drift or pipeline conflicts.
- **Unified Debug Window Layout:** Consolidates independent dashboard view paradigms into a multi-column single-viewport. Evaluating entities sit adjacent to action panels, producing synchronous visual tracking loops during functional audits.

Setup & Deployment Instructions

1. **Provision Virtual Mock Drivers:** Navigating through *Settings > Devices & Services > Helpers*, provision the following identical components:
 - `input_select.active_source` (Dropdown state selector)
 - `input_boolean.tv_power_state` (Virtual TV power toggle state)
 - `input_boolean.avr_power_state` (Virtual AVR power toggle state)
 - `input_number.media_volume` (Absolute volume level scale slider from 0 to 100)
2. **Instantiate Lovelace Frontend:** Add an empty manual card container component layout to your preferred workspace view, switch to the raw code input terminal modal, paste the provided YAML matrix block, and commit changes.
3. **Execute Performance Validation:** Take the dashboard out of configuration edit mode. Toggle power blocks, drag the volume bars, and switch active drop selections to evaluate synchronous event traces across all logging panes.